

Convergence Rules for Improper Integrals

Type I (Integrate to $\pm\infty$)

$$\int_1^{\infty} \frac{1}{x^p} dx \quad \text{Divergent iff } p \leq 1$$

Examples. $\int_1^{\infty} \frac{1}{x^2} dx$ **convergent**

$$\int_1^{\infty} \frac{1}{x} dx \quad \text{divergent}$$

$$\int_1^{\infty} \frac{1}{\sqrt{x}} dx \quad \text{divergent}$$

Type II (Integrate to **asymptote**)

$$\int_0^1 \frac{1}{x^p} dx \quad \text{Divergent iff } p \geq 1$$

Examples. $\int_0^1 \frac{1}{x^2} dx$ **divergent**

$$\int_0^1 \frac{1}{x} dx \quad \text{divergent}$$

$$\int_0^1 \frac{1}{\sqrt{x}} dx \quad \text{convergent}$$

Comparison Theorem

If $0 \leq f \leq g$ and $\begin{cases} \int f dx \text{ is divergent,} & \text{then } \int g dx \text{ is divergent} \\ \int g dx \text{ is convergent,} & \text{then } \int f dx \text{ is convergent} \end{cases}$

Example. $\int_{10}^{\infty} \frac{1}{x^3+2x+1} dx$

Compare to $\frac{1}{x^3}$.

Both are positive for $x \geq 10$.

Comparing to **bigger** function $\frac{1}{x^3+2x+1} < \frac{1}{x^3}$

Which has $\int_{10}^{\infty} \frac{1}{x^3} dx$ **convergent** ($p = 3 > 1$).

So $\int_{10}^{\infty} \frac{1}{x^3+2x+1} dx$ is **convergent** also.

Example. $\int_2^5 \frac{x}{(x-2)^3} dx$

Compare to $\frac{1}{(x-2)^3}$.

Both are positive for $2 < x < 5$.

Comparing to **smaller** function $\frac{1}{(x-2)^3} < \frac{x}{(x-2)^3}$

Which has $\int_2^5 \frac{1}{(x-2)^3} dx$ **divergent** ($p = 3 \geq 1$).

So $\int_2^5 \frac{x}{(x-2)^3} dx$ is **divergent** also.

1. Decide “Convergent” or “Divergent” for the following Type I Improper Integrals.
(DO NOT INTEGRATE)

(a) $\int_8^{\infty} \frac{5}{\sqrt[4]{x}} dx$

(c) $\int_{-\infty}^{-2} \frac{7}{x^3} dx$

(b) $\int_4^{\infty} \frac{2x+1}{x^2-x-3} dx$

(d) $\int_{-\infty}^{-7} \frac{2x^2-1}{x^4+1} dx$

2. Decide “Convergent” or “Divergent” for the following Type II Improper Integrals.
(DO NOT INTEGRATE)

(a) $\int_0^7 \frac{5}{\sqrt[4]{x}} dx$

(c) $\int_{-5}^0 \frac{7}{x^3} dx$

(b) $\int_1^3 \frac{x+5}{(x-1)(x+4)} dx$

(d) $\int_2^{12} \frac{x-1}{(x+1)(x-2)^2} dx$